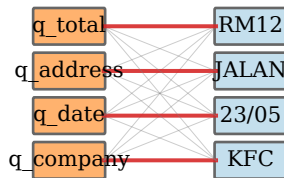


FOCUS assigner explained: basic → intermediate → advanced

Basic: what is the assigner? **Intermediate:** 4 learned queries **Advanced:** training objective

ASSIGNER
 OCR tokens (bag of words) → field values (company / date / ...)
 Goal: pick the right token for each field.



$$\alpha_{f,i} = \text{softmax}_i(q_f^\top Wk_i / \sqrt{d})$$

$$\mathcal{L}_{pos} = -\log \sum_{i \in P_f} \alpha_{f,i}$$

with text priors $\phi_i \in \mathbb{R}^6$ gating k_i .

⇒ multi-instance NLL + hierarchical L1 → L2

Why FOCUS: +0.24 F1 over Rule-based **Why FOCUS fails: high-entropy loss** **Hierarchical L1 → L2 routing**

